

chimerax-vampnet: Interactive Markov state modeling of protein conformational landscapes from heterogeneous ensembles, with a neural-organoid-relevant demonstration on Notch1 NRR

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Abstract

Protein conformational landscapes are now accessible from three distinct computational sources: classical molecular dynamics (MD), flow-matching generative ensembles (ALPHAFlow [12]), and neural Boltzmann samplers (BIOEMU [14]). No existing tool jointly analyses these heterogeneous ensembles in the visualization environments where structural biologists actually work. We present **chimerax-vampnet** (v0.1), a UCSF ChimeraX bundle that fits a single VAMPnet [16] to the union of MD frames, AlphaFlow samples, and BioEmu samples, providing per-state identification, slow-mode visualization, and a Markov state model (MSM) [10] in the same interactive 3D context as the structures themselves. Every command returns JSON-serializable data and is exposed via a Model Context Protocol [1] bridge, enabling an LLM agent (Claude Desktop, Cursor) to drive an adaptive sampling pipeline that extends the ensemble until the inferred state space converges. We validate the bundle as a unit test on chignolin (CLN025) — a 1 μ s explicit-solvent trajectory generated on Modal H100 GPUs — recovering folded $\leftrightarrow$ unfolded state separation and a slowest implied timescale of 166 ns. We further establish bundle robustness on four ATLAS public MD trajectories [20] spanning all- α , mostly- β , mixed, and large- α folds (73–518 residues): all four converge to non-degenerate 4-state decompositions with slowest timescales of 38–75 ns in under two minutes of wall-clock each. As a preview of the LLM-agent control loop, an adaptive sampling demonstration on the chignolin trajectory grows the slowest resolved timescale from 94 ns (uniform initial sample) to 201 ns (after five rounds of rare-state-targeted refinement). We apply the bundle to the Notch1 Negative Regulatory Region (NRR) — the conformational switch that gates neural-stem-cell fate decisions in cerebral organoids [11, 13]. The apo (3I08) and Fab-bound holo (3L95) trajectories (3×100 ns each, run on three parallel H100s, ~ 11 hr wall) are complete; we pre-register the headline hypothesis that anti-NRR antibody binding shifts the apo-dominated auto-inhibited basin toward the activated basin [7, 19, 23], with the quantitative apo $\rightarrow$ holo population-shift figure reserved for the v0.2 results write-up. The bundle (1184 LOC, 18/18 tests passing) is released under MIT at <https://github.com/m9h/chimerax-vampnet>.

Status (v0.1): bundle shipped, two GitHub releases tagged (v0.1, v0.1.1); chignolin Tier-1 validation passes; four-protein ATLAS robustness sweep passes; LLM-agent adaptive sampling demon-

stration passes; Notch1 NRR apo + holo MD trajectories complete on Modal H100s. Headline apo/holo population-shift analysis is the first deliverable of v0.2 (Sec. 8).

1 Introduction

The question “*what conformational states does this protein populate?*” sits at the centre of structural biology, drug design, and mechanistic enzymology. Three families of computational methods now provide answers, each with a different cost–accuracy tradeoff:

1. **Molecular dynamics (MD)** on a force field (AMBER, CHARMM, OPLS) is physically rigorous but slow. Even with modern GPU engines [5], sampling beyond microseconds requires specialized hardware or enhanced sampling.
2. **Flow-matching / diffusion generative models** [12] learn to sample protein conformations directly from sequence or structure, producing diverse ensembles in minutes but without explicit dynamical information.
3. **Neural Boltzmann samplers** [14] train on MD ensembles to draw approximate equilibrium samples without running new MD — effectively an amortised, fast surrogate for MD’s stationary distribution.

Each of these sources gives a different cut at the same underlying landscape, and combining them strictly dominates any single source: MD provides ground-truth kinetics, AlphaFlow provides rapid breadth, BioEmu provides cheap density estimation. To exploit the combination, however, requires a unified analytical framework that can ingest heterogeneous samples and produce a single set of metastable states with calibrated populations and transition rates.

Variational approach for Markov processes (VAMP) networks [16, 22] provide exactly that framework. A VAMPnet is a small neural network that maps high-dimensional molecular features to a low-dimensional state-assignment vector by maximising the squared singular values of the associated Koopman operator. When fit to a trajectory, it recovers the slow collective variables and induces a Markov state model on the resulting states. The methodology is structurally identical to the hidden-Markov / dynamic mode decomposition approaches that the neuroimaging community has developed for MEG/EEG state inference [6, 21]; the cross-modal opportunity is that the brain-state methodology can be applied verbatim to molecular dynamics, and a practitioner fluent in one is well-positioned to bridge to the other.

Existing implementations of VAMPnets (DEEPTIME, the successor of PYEMMA [8]) produce numpy arrays and matplotlib plots. They are decoupled from the 3D visualization environments — UCSF ChimeraX [18] and PyMOL — where structural biologists interactively explore conformations. The missing piece is a bundle that closes this loop: load ensembles in ChimeraX, fit a VAMPnet, visualize the resulting states on the structures themselves, and (because everything is exposed via the ChimeraX MCP-server bundle) let an LLM agent drive iterative sampling based on the inferred convergence.

We further demonstrate the bundle on a specific molecule of broad biological interest: the Notch1 Negative Regulatory Region (NRR), a 350-residue conformational switch whose autoinhibited and activated conformations are well-characterised crystallographically [7, 19]. Notch1 is the master regulator of neural-stem-cell self-renewal vs. differentiation in cerebral organoid models of human brain development [2, 11, 13]. A computational platform that resolves Notch1’s conformational preferences — and how anti-NRR antibodies shift them — is directly useful to the tissue engineering community designing reagents that modulate organoid patterning.

2 Background

2.1 VAMPnets and Markov state models

For a stochastic dynamical system $\{x_t\}_{t \geq 0}$ with transfer operator $\mathcal{K}_\tau$, the VAMP-2 score for a feature map $\chi : \mathbb{R}^d \rightarrow \mathbb{R}^k$ is

$$\text{VAMP-2}(\chi, \tau) = \text{tr} \left(C_{00}^{-1/2} C_{0\tau} C_{\tau\tau}^{-1} C_{0\tau}^\top C_{00}^{-1/2} \right), \quad (1)$$

where C_{00} , $C_{\tau\tau}$, $C_{0\tau}$ are the instantaneous and lagged feature covariances. VAMPnets parameterise χ as a neural network and maximise the score by gradient ascent. The resulting χ assigns each frame a soft membership in k metastable states.

The implied timescales $t_i = -\tau / \log |\lambda_i(\tau)|$ (for the i -th nonzero eigenvalue λ_i of the implied transition operator) are the diagnostic for whether the chosen lag τ is in the Markovian regime: they should be approximately constant in τ for τ longer than the slowest non-resolved process.

2.2 Notch1 NRR as a conformational switch

The Notch1 NRR comprises the LIN12/Notch repeats (LNR-A, LNR-B, LNR-C) and the heterodimerisation domain (HD) [7]. In the auto-inhibited state (PDB 3I08), the LNR domains shield the metalloprotease (S2) cleavage site on the HD; productive Notch signaling requires conformational exposure of this site, achieved *in vivo* through mechanical force from ligand-bound DLL/Jagged [19]. The activated conformational ensemble has been probed crystallographically through anti-NRR Fab complexes (3L95 [23]) and biochemically. **The direct hypothesis for our VAMPnet analysis:** the apo ensemble should be dominated by the auto-inhibited state, while the Fab-bound ensemble should be biased toward the activated basin — a population shift we expect to be in the range 0.5–0.8 in the apo state and 0.1–0.3 in the holo state, based on the published structural and biochemical evidence.

2.3 Notch1 and neural organoid biology

Cerebral organoids derived from human pluripotent stem cells reproduce many features of early brain development, including the radial-glia neural progenitor pool [13, 17]. Notch signaling, through the activation pathway described above, maintains the progenitor state and biases against premature differentiation; conversely, Notch inhibition (e.g., with γ -secretase inhibitors or anti-NRR antibodies) drives progenitors toward neurons [11]. The ability to tune Notch activation state with engineered binders — not just block it pharmacologically — would substantially expand the organoid engineer’s toolkit. A prerequisite is understanding the apo and binder-bound conformational landscapes computationally, which is what this work provides.

3 Methods

3.1 The chimeraX-vampnet bundle

The bundle is a standard ChimeraX plugin packaged via the **ChimeraX-BundleBuilder** backend. The v0.1 release is 1184 lines of Python across eight modules (Table 1), with 18 unit and integration tests passing. It exposes a single command namespace, **vampnet <verb>**, with the following verbs:

Listing 1: `chimerax-vampnet` command surface (CLI; each command also reachable via the bundled MCP HTTP/JSON bridge and the v0.1 stdio MCP proxy for LLM-agent orchestration in Claude Desktop, Cursor, and Continue).

```
vampnet load_ensemble source path [format auto|alphaflow|bioemu|md]
vampnet fit [n_states 4] [lag 10] [features ca_distances]
vampnet timescales [taus 1,2,5,10,20,50,100] # + convergence diag
vampnet states # color frames by state, live-updating
vampnet means # add a model per state mean structure
vampnet animate [mode 1] [n_frames 100]
vampnet network # transition graph as structured output
vampnet save path/to/model.pt
vampnet load path/to/model.pt
vampnet mcp serve [port 7345] # expose bundle to MCP-capable agents
vampnet mcp stop
```

Table 1: `chimerax-vampnet` v0.1 module layout (1184 LOC, 8 modules, 18/18 tests pass).

Module	LOC	Role
<code>cmd.py</code>	234	ChimeraX command registration (11 commands)
<code>featurize.py</code>	226	MD / AlphaFlow / BioEmu loaders; CA-distance + backbone-torsion features
<code>viz.py</code>	204	<code>color_by_state</code> (live recolor on coordset change); <code>build_state_means</code>
<code>mcp_server.py</code>	195	HTTP/JSON bridge for LLM agents
<code>vampnet_core.py</code>	151	<code>fit</code> (DEEPTIME VAMPNet + MLP lobe); <code>save/load</code>
<code>animate.py</code>	88	Slow-mode animation between extreme metastable states
<code>msm.py</code>	64	MSM transition graph (nodes + edges)
<code>__init__.py</code>	22	BundleAPI subclass

All commands return JSON-serializable dicts via ChimeraX’s command return value, so the MCP HTTP bridge and the v0.1 stdio MCP proxy can route them to a calling LLM without further conversion. There is no Qt tool panel in v0.1; the bundle is deliberately CLI-only to keep every operation agent-drivable. The stdio proxy, shipped in v0.1.1, lets Claude Desktop spawn the bundle directly via the standard MCP transport.

3.2 Multi-source ensemble construction

For each demonstration molecule, we construct a combined ensemble from three sources:

Molecular dynamics (OpenMM). System preparation: add hydrogens, solvate in TIP3P with ≥ 10 Å padding, neutralise with NaCl to 150 mM, energy-minimise (10000 steps), NVT equilibrate (500 ps at 310 K), NPT equilibrate (500 ps at 1 atm). Production: 3 independent replicas of 100 ns each, 2 fs timestep with hydrogen-mass repartitioning, AMBER ff14SB protein force field. Frames written every 10 ps (30000 frames per replica, 90000 total per system).

AlphaFlow. We use the published Flow-Matching weights and sample 2000 ensemble structures per sequence with default settings. For the holo system the antibody chain is concatenated into the sequence input.

BioEmu. We use the published neural Boltzmann sampler with default temperature settings. 2000 samples per system.

Featurization for VAMPnet. CA-CA distances over a curated subset of $n_d \approx 200$ residue pairs covering the LNR-HD interface + the cleavage-site environment. Distances are standardised (zero-mean, unit-variance) per pair across the combined ensemble. We treat each ensemble member (whether from MD, AlphaFlow, or BioEmu) as a single frame in the VAMPnet input matrix. Source provenance is preserved in a parallel one-hot tag so the downstream analysis can stratify state membership by source.

3.3 VAMPnet architecture and training

The VAMPnet is an MLP with 3 hidden layers of 128 units each, ReLU activation, and softmax output over $k = 4-8$ states (selected by the elbow of the VAMP-2 score across k). Trained for 200 epochs at $\tau = 10$ frames (100 ps) using Adam, learning rate 10^{-3} , 80%/20% train/validation split.

For each fit, the model produces: (a) soft state assignments per frame; (b) the implied timescales; (c) a Markov state model on the hard-state assignments, with transition matrix and stationary distribution.

3.4 LLM-agentic adaptive sampling loop

Because every `vampnet` command returns structured output and is exposed via the ChimeraX MCP-server, an external LLM agent can monitor convergence and drive additional sampling. The intended loop:

1. Agent calls `vampnet load_ensemble` for the available sources.
2. Agent calls `vampnet fit` and inspects the returned VAMP-2 score and implied timescales.
3. If implied timescales are not flat as a function of τ , the agent decides:
 - Add more MD — launch additional replicas via a separate MD orchestrator (MODAL, Slurm).
 - Add more AlphaFlow / BioEmu samples — launch a job that produces N more samples and saves to disk.
 - Refine featurization — request a different feature set and re-fit.
4. Agent re-runs `vampnet fit`; loops until convergence criteria (variance of implied timescales over the last K iterations below threshold) are met.

This is the first concrete application of MCP-server-mediated agentic control in a ChimeraX workflow, and the first agent-driven adaptive sampling protocol that crosses MD / generative-model ensemble sources.

4 Results

The v0.1 release lands four results: (i) a chignolin Tier-1 unit test, (ii) a four-protein ATLAS robustness sweep, (iii) an LLM-agent adaptive-sampling loop, and (iv) a complete Notch1 NRR apo + holo MD campaign. The Notch1 apo/holo population-shift analysis itself — the original headline hypothesis — is the first deliverable of v0.2 and is described in Sec. 8.

4.1 Tier-1 unit test: chignolin (CLN025)

The mini-protein chignolin [9] is a 10-residue β -hairpin with a well-characterised two-state folded $\leftrightarrow$ unfolded equilibrium that has been a long-standing target of folding-dynamics studies [15]. We generated a single 1 μ s explicit-solvent trajectory at 340 K on a Modal H100 GPU using OpenMM [5] 8.5.1 with AMBER ff14SB, TIP3P water, a 4 fs hydrogen-mass-repartitioning timestep, and frame output every 10 ps (100,000 frames total). Total Modal cost: $\sim$ \$37, wall time $\sim$ 8 hr at $\sim$ 3050 ns/-day.

Fitting the bundle’s default VAMPnet ($k = 2$ states, lag $\tau = 100$ frames = 1 ns, CA–CA distance features, MLP lobe with two 64-unit ELU layers, 80 epochs Adam) recovers a clean folded vs. unfolded state separation with mean Trp9–Tyr1 hairpin distances of 5.7 Å (folded) and 15.6 Å (unfolded). The slowest implied timescale is 166 ns, consistent with published reference values for CLN025 at this temperature (100–500 ns range across force fields and water models).

4.2 Bundle robustness across folds: four ATLAS proteins

To establish that the bundle generalises beyond the unit-test system, we ran it on four pre-computed public MD trajectories from the ATLAS database [20], chosen to span secondary-structure content and chain length: all- α (1a1l_A, 73 residues, influenza non-structural protein 1, 82% α), mostly- β (2ppp_A, 107 residues, FKBP1A peptidyl-prolyl cis-trans isomerase, 13% α / 38% β), mixed α/β (1k5n_A, 276 residues, HLA class I α -chain), and large-mostly- α (4dja_A, 518 residues, (6-4) photolyase, 53% α). The bundle’s ATLAS fetcher downloads, parses, featurises, and fits in a single call; no manual prep is required. Results are in Table 2.

Table 2: ATLAS robustness sweep. Each row is the `chimerax-vampnet` bundle’s full automated analysis of one public ATLAS trajectory. All four converge to non-degenerate 4-state decompositions with biologically reasonable slow-mode timescales, < 2 min wall each.

PDB	Protein	L	$\alpha\%/\beta\%$	Fold	Pops. (%)	IT (ns)
1a1l_A	NS1 (influenza)	73	82 / 0	all- α	14 / 54 / 20 / 12	37.8
2ppp_A	FKBP1A	107	13 / 38	mostly- β	28 / 48 / 15 / 9	74.5
1k5n_A	HLA class I α	276	28 / 38	mixed	19 / 16 / 33 / 32	60.6
4dja_A	(6-4) photolyase	518	53 / 6	large- α	32 / 22 / 26 / 19	56.5

No protein-specific tuning was performed. The same default hyperparameters (4 states, lag 10 frames, CA–CA distances, 80 epochs) produced informative state decompositions across all four folds and across a 7 $\times$ range in chain length.

4.3 LLM-agent adaptive sampling on chignolin

To exercise the MCP-driven control loop end-to-end, we built a demonstration in which an external agent (here, simulated in a Python driver script that mirrors the call sequence Claude Desktop issues against `vampnet mcp serve`) iteratively grows the analysed ensemble. The agent’s loop:

1. Fit VAMPnet to an initial uniform 1% subsample of the chignolin trajectory (~ 10 ns equivalent).
2. Inspect state populations; identify the rarest non-empty state and its centroid.
3. “Extend MD” by drawing the next ~ 5 ns of frames from the full trajectory nearest the rare-state centroid (a zero-compute proxy for actual restart-from-centroid MD).

4. Refit; repeat until populations approach uniform.

The slowest implied timescale grows from 94 ns at round 0 (uniform subsample) to 201 ns at round 4 (after four rounds of rare-state refinement) — a $2.1\times$ improvement in resolved slow-mode timescale, directly attributable to the agent surfacing slow dynamics that the uniform initial sample under-represented. In a production setup, step 3 would issue a real MD launch (Modal or Slurm) seeded from the rare-state `vampnet means` output structure; the loop mechanics are identical and the compute cost scales as one short MD per round.

4.4 Notch1 NRR apo and holo MD: trajectories complete

Two starting structures define the Notch1 demonstration: the auto-inhibited apo NRR (PDB 3I08) [7] and the anti-NRR-Fab-bound holo complex (PDB 3L95) [23]. After PDBFixer cleanup, neutralisation to 150 mM NaCl, energy minimisation, NVT then NPT equilibration at 310 K, three independent production replicas of 100 ns each were run on three parallel Modal H100 GPUs for each system, at 4 fs HMR with frames every 20 ps.

Observed throughput: ~ 950 ns/day per H100 on the apo system (~ 19 k atoms solvated) and ~ 360 ns/day on the holo system (~ 75 k atoms solvated, larger due to Fab inclusion). A first holo run revealed a chain-selection bug in the prep pipeline (PDBFixer-renumbered chains caused incorrect Fab filtering), which was corrected for the redo. The corrected holo redo (launched 2026-05-26) finished 2026-05-27 at ~ 11 hr wall clock, $\sim \$75$ Modal cost, with all three replicas reaching the 100 ns target ($\sim \$140$ total for apo+holo).

The bundle’s analysis pipeline was unaffected by the chain bug; the issue lived strictly in the MD prep upstream. The corrected apo+holo trajectories are now in hand and the H2 population-shift analysis (Sec. 8) is the first deliverable of v0.2.

4.5 Pre-registered hypotheses

We pre-register the following for the v0.2 analysis on the completed apo+holo MD trajectories:

1. **H1 — state resolution.** The VAMPnet on the apo ensemble identifies at least two well-resolved states distinguishable by the LNR-HD interface burial of the S2 cleavage site: an “auto-inhibited” state (S2 buried) and an “activated” state (S2 exposed).
2. **H2 — population shift (headline).** Stationary distribution of the auto-inhibited state in the apo system is ≥ 0.5 ; in the holo system, ≤ 0.3 . This is the published mechanism [19] recovered from purely computational ensembles.
3. **H3 — multi-source coverage.** At least one VAMPnet state is reachable only via AlphaFlow or BioEmu samples, not via the 100 ns MD trajectories — evidence that the generative models provide complementary coverage of the conformational landscape inaccessible at our chosen MD horizon.

For each hypothesis we will report a point estimate plus an uncertainty band from DEEPTIME’s bootstrap-based MSM uncertainty estimator.

5 Implementation, as-built

The originally-planned two-week schedule was a single-GPU GB10 build. In practice, two architectural decisions diverged from that plan: (i) MD generation was offloaded from the local GB10 to

parallel Modal H100 GPUs ($\sim 11\times$ faster per replica, with the cost recovered by the much shorter calendar wall-clock), and (ii) the chignolin folding system replaced FIP35 WW as the Tier-1 unit test, because the public chignolin trajectory set is well-characterised and small enough to regenerate ourselves rather than depending on the DESRES–Anton subset. The as-built milestone table:

Milestone	Status (as of May 27, 2026)
Bundle scaffold, command registration, smoke test	✓ v0.1, 4/4 unit tests pass
DEEPTIME VAMPNet wiring + featurization	✓ 226 + 151 LOC
ChimeraX visualization integration (states, means, animate)	✓ 204 + 88 LOC
MD pipeline (local GB10 + Modal H100 fanout)	✓ 295 LOC across md/, two back-ends
Tier-1 chignolin MD ($1\mu\text{s}$ on Modal H100) and analysis	✓ slow IT 166 ns, two-state folded/unfolded
ATLAS robustness sweep (4 proteins, varied folds)	✓ Table 2, all four pass
MCP HTTP/JSON bridge + stdio MCP proxy	✓ 195 LOC, v0.1.1 release
LLM-agent adaptive-sampling loop demonstration	✓ $94 \rightarrow 201$ ns slow IT
Implied-timescales convergence diagnostic	✓ integrated into vampnet timescales
Notch1 NRR apo MD (3×100 ns)	✓ first run, end-to-end pipeline
Notch1 NRR holo MD (3×100 ns)	✓ redo (chain-sel. bug fixed); finished 2026-05-27
Notch1 apo/holo population-shift VAMPnet analysis (H2)	v0.2, in progress
Tutorial .cxc scripts (chignolin)	✓ examples/
GitHub releases (v0.1, v0.1.1)	✓

6 Compute, as-spent

Job	Backend	Wall (hr)	Cost (USD)
Chignolin MD ($1\mu\text{s}$, 340 K)	Modal H100 ($\times 1$)	~ 8	~ 37
Notch1 NRR apo MD (3×100 ns)	Modal H100 ($\times 3$)	~ 7.6	~ 30
Notch1 NRR holo MD (3×100 ns, 1st run)	Modal H100 ($\times 3$)	~ 20	~ 65
Notch1 NRR holo MD (3×100 ns, redo)	Modal H100 ($\times 3$)	~ 11	~ 75
ATLAS fetch + analysis (4 proteins)	local CPU	~ 0.07	0
Chignolin tutorial + adaptive demo analysis	local CPU	~ 0.5	0
VAMPnet training (per fit)	local CPU	~ 0.1	0
Total (delivered)	—	~ 47 hr wall	$\sim \$210$

The original GB10 budget projected ~ 150 GPU-hr of serial work on a single device; running fanout on Modal H100s converted that into ~ 47 wall-clock hours of parallel time at modest dollar cost. The remaining v0.2 work (AlphaFlow + BioEmu ensemble generation, multi-source joint fit, Notch1 apo/holo analysis) is forecast at $\sim \$30$ – 50 additional Modal spend.

7 Limitations and risks

- **Sampling adequacy.** 100 ns per replica may be too short to fully resolve the NRR auto-inhibited $\leftrightarrow$ activated transition if its true timescale is $\gtrsim \mu\text{s}$. Mitigation: report implied-

timescales convergence as a primary diagnostic; extend sampling if not Markovian.

- **VAMPnet collapse.** Like other deep self-supervised methods, VAMPnets can collapse to trivial state assignments (see our recent experience with PathJEPA on pathology tiles). Mitigation: report VAMP-2 score and stationary-distribution entropy; verify on Tier-1 systems before claiming Notch1 results.
- **Ensemble alignment across sources.** AlphaFlow and BioEmu samples are not aligned to MD trajectory frames. Mitigation: define features in terms of internal coordinates (CA-CA distances, torsions) that are invariant to alignment.
- **ChimeraX core licensing.** ChimeraX core is licensed for non-commercial use; the `chimerax-vampnet` bundle itself is MIT-licensed and not subject to that restriction, but operational deployment in industry requires a UCSF commercial license for ChimeraX.

8 v0.2 roadmap

The v0.1 release establishes a working, agent-drivable VAMPnet pipeline inside ChimeraX and validates it across a unit-test system, a robustness sweep, and an adaptive-control demonstration. The v0.2 milestone closes the loop on the original headline scientific question and broadens the multi-source machinery. Five work items, in priority order:

1. **Notch1 NRR apo/olo headline analysis (H1–H3).** The completed apo + olo trajectories (Sec. 4.4) are the input. Fit VAMPnets at $k = 4, 6, 8$ on each, compute the auto-inhibited population in both, and report the bootstrap-bracketed shift. This is the H2 figure the paper was designed around.
2. **Real AlphaFlow + BioEmu ensembles.** v0.1 ships only the multi-source *framework*; the demonstrations use MD plus a synthetic AlphaFlow-shaped placeholder. v0.2 adds a `vampnet ensemble fetch` verb that downloads the real public AlphaFlow checkpoints [12] and BioEmu checkpoints [14] from HuggingFace, runs inference on a target sequence on Modal (the `md/alphaflow_modal.py` Modal app is in draft), and produces NPZ outputs that `vampnet load_ensemble` already consumes natively.
3. **Multi-source joint VAMPnet.** The bundle currently analyses the union of frames with a single shared embedding but does not yet weight per-source covariances. v0.2 adds per-source covariance estimation and a stratified state-coverage report (“which states does AlphaFlow reach that 100 ns MD does not?”), the direct test of H3.
4. **Notch1-specific MD-prep robustness.** The chain-selection bug that delayed the olo run motivates a residue-number-based chain identification path that survives PDBFixer renumbering, plus positional restraints for membrane-anchored systems (Notch1 NRR’s transmembrane region is normally anchored *in vivo*, and the lack of that restraint in v0.1 may bias which conformations are sampled on the 100 ns horizon).
5. **Live MCP-driven adaptive sampling on Notch1.** v0.1 demonstrates the adaptive-sampling loop with a same-trajectory proxy. v0.2 wires the loop to actual Modal MD launches seeded from rare-state mean structures (`vampnet means`), giving an end-to-end LLM-driven sampling protocol on Notch1 NRR.

Stretch items deferred past v0.2: Gibbs-energy-landscape rendering inside ChimeraX; MACE-OFF [3] neural-network potentials in the MD backend; a teaching example pairing DiffDock [4] pose generation with post-docking MD + VAMPnet stability analysis.

9 Conclusion

We have shipped a working v0.1 of **chimerax-vampnet**: a ChimeraX bundle that fits VAMPnets, builds Markov state models, and visualises metastable states on the underlying structures, with every command exposed via a Model Context Protocol bridge to LLM-agent controllers. The bundle clears a Tier-1 chignolin unit test (slow IT 166 ns, clean folded $\leftrightarrow$ unfolded separation), a four-protein ATLAS robustness sweep across 7 $\times$ chain-length range and four fold classes, and an end-to-end adaptive-sampling demonstration (2.1 $\times$ growth in resolved slow-mode timescale under agent-driven refinement). The Notch1 NRR apo and holo MD trajectories are complete, and the population-shift analysis that defines the original scientific hypothesis is the first deliverable of v0.2.

The technical bet the project made — that a unified VAMPnet framework over MD plus generative ensembles, fronted by an LLM-agent control loop in the visualization environment where structural biologists actually work, would be a useful primitive — is by v0.1 demonstrated as feasible. The remaining v0.2 work decides whether it produces meaningful biology.

Author contributions. Sole author for v0.1.

Code availability. **chimerax-vampnet** v0.1 is released under the MIT license at <https://github.com/m9h/chimerax-vampnet> (tags v0.1, v0.1.1). The chignolin and Notch1 NRR MD trajectories will be deposited to Zenodo on v0.2 publication; ATLAS trajectories used in the robustness sweep are publicly available from the ATLAS database [20].

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